QUESTION 1

Which of the following are true about RAM in embedded systems?

Answer is Wrong **Score: 0 / 1**

Options

Wrong

- [x] A. RAM is non-volatile

Expected

- [ ] B. RAM provides constant time access

Expected

- [ ] C. RAM loses data when powered off

- [ ] D. RAM can only be read, not written

FEEDBACK

Solution: RAM allows random (constant-time) access and is volatile, meaning it loses data when the system is turned off. Link: https://youtu.be/_KwtU1EzPS0?feature=shared&t=204

QUESTION 2

What is the key difference between Flash memory and RAM?

Not Attempted **Score: 0 / 1**

Options

A. Flash is faster than RAM

Expected

B. Flash is non-volatile, RAM is volatile

C. Flash cannot be used to store code

D. RAM retains data without power

FEEDBACK

Solution: Flash is non-volatile and retains data after power is removed, whereas RAM is volatile.

Link: https://youtu.be/_KwtU1EzPS0?feature=shared&t=525

QUESTION 3

What are the advantages of static memory allocation in embedded systems?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Reduced memory fragmentation

B. Memory allocated only at runtime

Expected

C. Predictable memory usage

Expected

D. No memory allocation failures at runtime

FEEDBACK

Solution: Static allocation occurs at compile time, ensuring predictability, reduced fragmentation, and avoids runtime failures.

Link: https://youtu.be/_KwtU1EzPS0?feature=shared&t=862

QUESTION 4

What are potential drawbacks of using dynamic memory allocation in embedded systems?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Can cause fragmentation

B. Ensures deterministic behavior

Expected

C. Memory leaks can occur

D. Unused memory is automatically freed

FEEDBACK

Solution: Dynamic allocation risks fragmentation and memory leaks, causing unpredictability and possible system crashes. Link: https://youtu.be/_KwtU1EzPS0?feature=shared&t=1441

QUESTION 5

Which techniques help optimize memory usage in embedded systems?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Use smallest suitable data types

B. Prefer global variables over local ones

Expected

C. Use data packing (e.g., RGB bits)

D. Prefer 32-bit integers for all data

FEEDBACK

Solution: Using optimal data types and packing values (like RGB in 16 bits) reduces memory waste and improves efficiency. Link: https://youtu.be/_KwtU1EzPS0?feature=shared&t=1621